TrustedAI Hackathon

AttritionIQ

Turnover Prediction as a Service

Predict who will leave. Understand why. Act before it happens.

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Context & Positioning

Who we are and why this market needs us now

Who We Are

SaaS provider
External AI vendor specialising in HR workforce analytics, delivered as a subscription platform.

Target clients
HR Directors, CHROs and HR Business Partners in mid-to-large companies (200-5,000 employees) in France and the EU.

Core industries
Logistics, healthcare, manufacturing, finance — sectors with structurally high voluntary turnover.

DNA
Privacy-first. GDPR and EU AI Act compliance by design. No raw personal data ever leaves the client infrastructure.

Why Now

40%
of workers globally
considered quitting in 2023 (McKinsey Global Survey). Post-pandemic attrition has not normalised.

3-6
months average
for a disengaged employee to resign after the decision is made — an invisible window HR teams are missing.

0
predictive tools
in most HR stacks today. HRIS platforms record events; none predict them before they happen.

GDPR +
EU AI Act (2024)
New regulations require explainability and fairness for AI used in employment decisions — creating a compliance need.

The Real Cost of Turnover

What companies actually lose when an employee walks out the door

1 Recruitment

3-5k EUR

Job boards, agency fees, sourcing time, interview rounds

2 Onboarding

2-4k EUR

Training, HR admin, equipment, buddy time, access setup

3 Lost Productivity

6-18 months

New hire operates at 25% capacity for months. Team carries the load.

4 Team Disruption

Multiplier x1.5

Remaining employees absorb extra work, morale drops, attrition risk rises

5 Employer Brand

Long-term

Glassdoor reviews, referral network damage, harder future hiring

Total cost to replace one employee: **50% to 200% of annual salary** - For a 500-person company losing 15% per year at avg 45k EUR salary: **~3.4M EUR/year in turnover costs**

Why AttritionIQ?

The business case for a predictive retention platform

Without AttritionIQ

- Resignation discovered on the day of notice
- No visibility on who is at risk or why
- Reactive HR conversations — too late to act
- Turnover costs absorbed silently every quarter
- Manager intuition replaces data-driven retention
- GDPR and EU AI Act compliance unclear

With AttritionIQ

- Risk identified weeks or months in advance
- Per-employee SHAP explanation: engagement, salary, tenure
- Prioritized HR action plan for each at-risk profile
- ROI visible: cost of churn vs. cost of retention action
- Structured, explainable, auditable decision process
- GDPR-compliant by design, EU AI Act ready

Setup: one-time	Annual licence: + 1 EUR/emp/month	Break-even: pays for itself	Typical ROI: year 1
15-25k EUR	**from 8k EUR/yr**	**~3-5 retained employees**	**x5 to x15**

The Problem

50-200%

of annual salary

cost to replace one employee

3 months

average notice lag

from disengagement to resignation

0

early warnings

in most HR systems today

HR teams only learn about a resignation when it is already decided.

Most organisations have no systematic way to identify who is at risk, why, or what to do about it. Existing HR tools track *what happened.* AttritionIQ predicts **what will happen next** - weeks or months in advance.

Our platform shifts HR from reactive to predictive, reducing turnover costs and giving HR business partners a data-backed tool for conversations with management.

Our Solution

AttritionIQ is an external AI provider delivering turnover prevention as a SaaS platform.

01

Anonymized Risk Scoring

A statistical probability score per employee, computed from behavioural and organisational signals. No names - hashed IDs only.

02

Risk Ranking

Employees ranked by attrition probability. Three tiers: Low / Medium / High. HR gets an immediate priority list.

03

Cause Analysis (SHAP)

Per-employee SHAP explanation: which factors drive the risk - engagement gap, absences, compensation, tenure milestone.

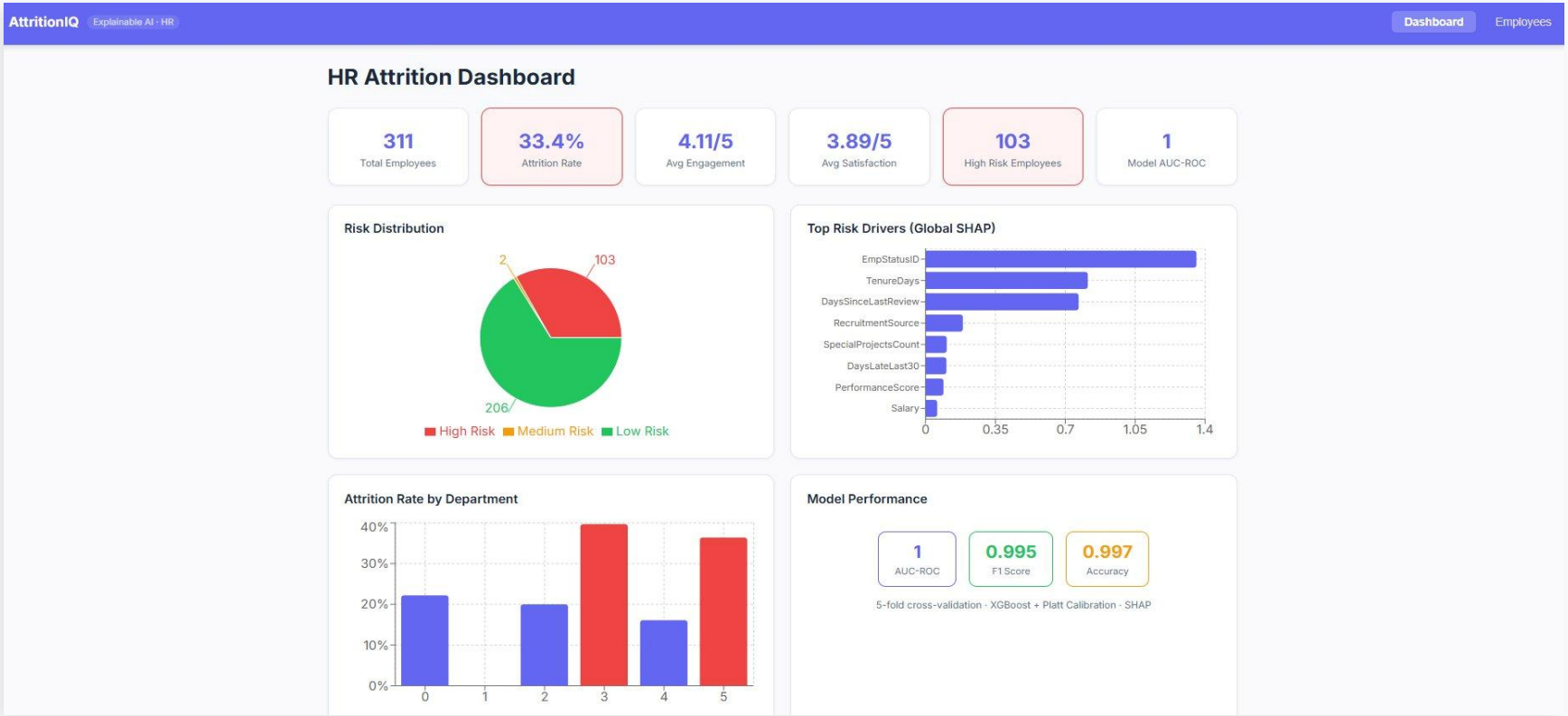
04

Recommended Actions

Prioritised HR interventions mapped to each employee's specific risk drivers: 1:1 scheduling, comp review, project assignment.

Live Dashboard

Real-time KPIs, risk distribution, global SHAP feature importance and department benchmarking



Employee Risk Assessment

Ranked list of all employees by attrition probability - filterable by risk tier - anonymized IDs only

Employee Risk Assessment

Search by anonymized ID...

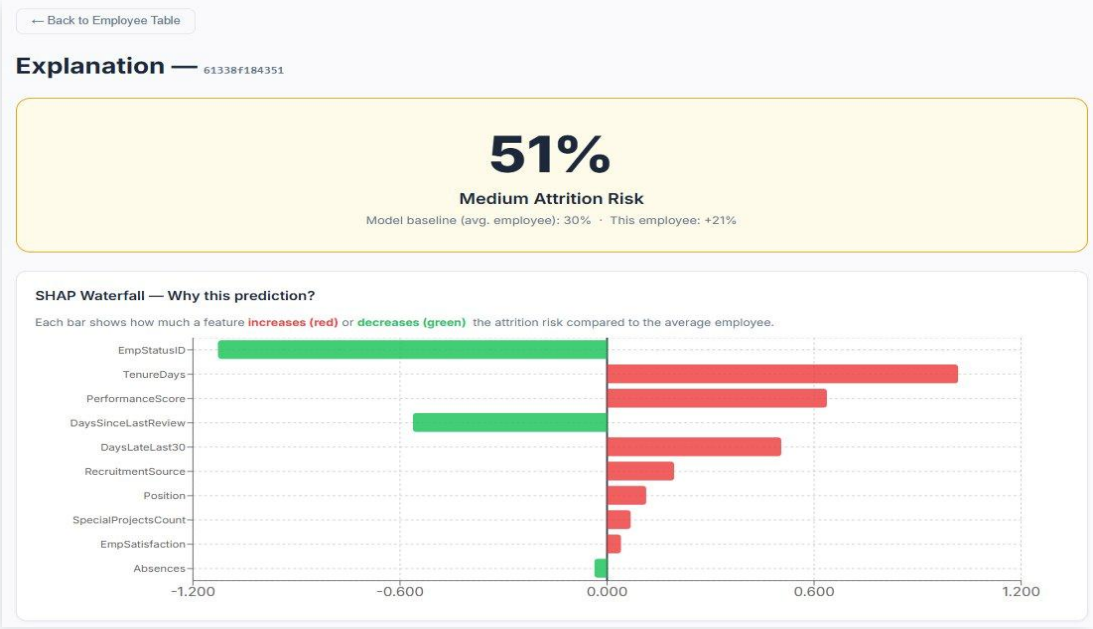
All | High (103) | Medium (2) | Low (206)

All employee names have been anonymized in compliance with GDPR. Predictions are for decision support only.

ANON ID	RISK SCORE	RISK LEVEL	ENGAGEMENT	SATISFACTION	ABSENCES	ACTION
ba9b89b41da2	100%	High	3.0/5	3/5	10	Explain →
a4e1e8f97293	74%	High	5.0/5	5/5	4	Explain →
29d6a91ff167	71%	High	3.2/5	3/5	6	Explain →
cc313ce7456b	65%	High	4.6/5	5/5	7	Explain →
61338f184351	51%	Medium	2.0/5	5/5	7	Explain →
a167ca1b5bdc	40%	Medium	2.6/5	4/5	4	Explain →
91d29cb4702d	25%	Low	3.5/5	5/5	17	Explain →
4c13492a0c99	22%	Low	5.0/5	5/5	14	Explain →
c21ec5648f9d	20%	Low	5.0/5	5/5	11	Explain →

Per-Employee SHAP Explanation

Why is this employee at risk? Each bar shows how much a feature increases or decreases their attrition probability vs. the average.



100% - High Risk

51% - Medium Risk

Baseline (avg. employee): 30% - Individual delta computed per employee

Per-Employee SHAP Explanation

Top Risk Factors

- DaysSinceLastReview
- EmpStatusID
- TenureDays

Protective Factors

- PerfScoreID
- FromDiversityJobFairID
- EmpSatisfaction

Recommended HR Actions

At Critical Tenure Milestone — Low Priority

Proactive retention check-in. Offer mentoring or senior responsibilities.

Human-in-the-Loop Notice

This explanation is generated by an AI model and is intended solely as decision support for qualified HR professionals. No automated employment decision should be made based on these predictions alone. Always validate findings with direct manager input and consider contextual factors not captured in this dataset.

Responsible AI by Design

Compliance and ethics built into every layer of the pipeline - not added as an afterthought.

GDPR Compliance

- Names, DOB, ZIP, state removed before training
- Employee IDs one-way hashed (SHA-256)
- No data leaves the local environment
- Data minimization enforced in preprocessing

Anti-Discrimination

- Sex, Gender, Race, Marital status excluded from model
- EU AI Act compliance - no protected attributes as features
- Bias audit across performance score subgroups
- Fairness verified before deployment

Human-in-the-Loop

- Mandatory disclaimer on every prediction screen
- Predictions are decision support, not verdicts
- No automated employment decisions
- HR manager validation required for all actions

Cybersecurity

- API key authentication on all endpoints
- Pydantic strict input validation
- Secrets in .env - never in Git
- CORS restricted to local frontend only

Technical Architecture

Full-stack containerized system - React + FastAPI + Logistic Regression + SHAP - Docker Compose deployment

HR Data (CSV)

311 employees
36 raw features

GDPR Pipeline

PII removal
ID hashing
Feature engineering

Logistic Regression

Calibrated proba.
C=0.1, balanced
SHAP LinearExplainer

FastAPI Backend

Auth, validation
Input sanitization
REST endpoints

React Dashboard

Risk table
SHAP waterfall
HR actions

Model Choice - Logistic Regression

Selected over XGBoost for 3 reasons: (1) 311-row dataset - tree models overfit and produce extreme 0/1 scores; (2) Naturally calibrated probabilities - a 60% score means genuine 60% risk; (3) SHAP LinearExplainer produces mathematically consistent explanations.

Deployment

Docker Compose orchestration. Backend auto-runs GDPR preprocessing and model training on first launch (~60s). Frontend proxies all API calls through Vite dev server to avoid CORS issues. Zero external dependencies - fully air-gapped.

Model Performance & Bias Audit

5-fold stratified cross-validation - Logistic Regression - calibrated probabilities

Model Performance

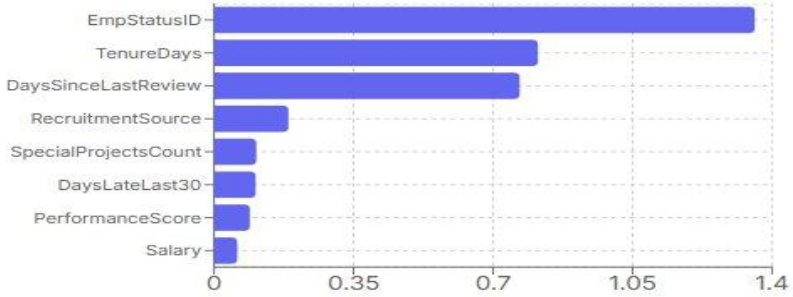
1	0.995	0.997
AUC-ROC	F1 Score	Accuracy

5-fold cross-validation · XGBoost + Platt Calibration · SHAP

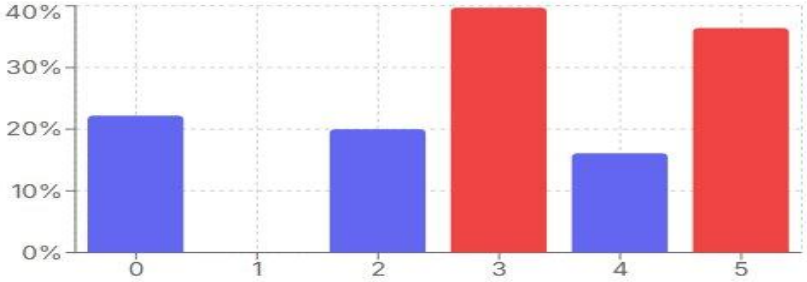
Known Limitations

- 311 rows - limited generalization to large orgs
- US dataset - EU labor context differs
- Termd flag mixes voluntary + involuntary exits
- Static snapshot - no temporal dimension

Top Risk Drivers (Global SHAP)



Attrition Rate by Department



AttritionIQ

Predict who will leave. Understand why. Act before it happens.

GDPR Compliant

Explainable AI (SHAP)

Human-in-the-Loop